Lab Report – EE-170L Computer Programming (Lab 02)

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1. Objectives

- Understand what a two-dimensional array is.
- Learn when to use two-dimensional arrays.
- Practice declaring and initializing 2D arrays.
- Access elements using row and column indices.
- Perform operations such as summing rows/columns and finding maximum values.

2. Theory

A **two-dimensional array** is the simplest form of a multidimensional array. It can be thought of as a table with rows and columns, where each element is identified by two indices: one for the row and one for the column.

Declaration Syntax:

```
type arrayName[rows][columns];
```

For example:

```
int matrix[3][3];
```

This declares a 3x3 integer array.

Initialization:

Arrays can be initialized row by row using nested braces, or in a single list. For example:

```
int a[3][4] = { {0,1,2,3}, {4,5,6,7}, {8,9,10,11} };
```

Accessing Elements:

Each element is accessed using two indices:

```
int val = a[2][3];      // element at row 2, column 3
```

Two-dimensional arrays are widely used in applications such as matrices, tables, and grids. They form the basis for advanced concepts like image processing, dynamic programming, and scientific simulations.

3. Lab Tasks and Programs

Task 1: Two-Dimensional Array Declaration and Initialization

```
#include <iostream>          // include input-output library
using namespace std;         // use standard namespace
int main() {                 // main function starts
int matrix[3][3] = { {1,2,3}, {4,5,6}, {7,8,9} };   // declare and
initialize 3x3 array
for(int i = 0; i < 3; i++) {      // loop for rows
for(int j = 0; j < 3; j++) {      // loop for columns
```

```
cout << matrix[i][j] << " ";    // print each element
}
cout << endl;                   // move to next line after each row
}
return 0;                       // end program
}
```

Task 2: Sum of Rows and Columns in a Two-Dimensional Array

```
#include <iostream>             // include input-output library
using namespace std;            // use standard namespace
int main() {                    // main function starts
 int matrix[3][3] = { {2,4,6}, {1,3,5}, {7,8,9} };    // declare and initialize 3x3 array
   int totalSum = 0;                          // variable for total sum
  for(int i = 0; i < 3; i++) {                // loop for rows
   int rowSum = 0;                            // sum of current row
   for(int j = 0; j < 3; j++) {               // loop for columns
   rowSum += matrix[i][j];                    // add element to row sum
    totalSum += matrix[i][j];                 // add element to total sum
  }
```

```
cout << "Sum of row " << i << " = " << rowSum << endl; //display row sum
 }
for(int j = 0; j < 3; j++) {          // loop for columns
int colSum = 0;                       // sum of current column
 for(int i = 0; i < 3; i++) {        // loop for rows
 colSum += matrix[i][j];       // add element to column sum
 }
 cout << "Sum of column " << j << " = " << colSum << endl; // display column sum
 }
cout << "Total sum = " << totalSum << endl;  // display total sum
return 0;                          // end program
}
```

Task 3: Finding the Maximum Value in a Two-Dimensional Array

```
#include <iostream>          // include input-output library
using namespace std;         // use standard namespace
int main() {                 // main function starts
```

```
int matrix[4][4] = { {1,9,3,7}, {4,2,6,8}, {5,11,13,0}, {12,14,10,15} };
// declare and initialize 4x4 array

int maxVal = matrix[0][0];      // assume first element is
maximum

for(int i = 0; i < 4; i++) {          // loop for rows

for(int j = 0; j < 4; j++) {          // loop for columns

if(matrix[i][j] > maxVal) {      // check if current element is
greater

maxVal = matrix[i][j];           // update maximum value

      }

   }

 }

cout << "Maximum value in array = " << maxVal << endl; //
display maximum value

return 0;                        // end program

}
```

4. Results

- Task 1 displayed the initialized 3x3 matrix with values from 1 to 9.

- Task 2 correctly calculated and displayed the sum of each row, each column, and the total sum of all elements.
- Task 3 successfully identified and displayed the maximum value in the 4x4 array.

These results confirm that two-dimensional arrays can store tabular data and allow operations like summation and searching for maximum values.

5. Conclusion

In this lab, we explored two-dimensional arrays in C++. We learned how to declare, initialize, and access elements using row and column indices. We practiced operations such as summing rows and columns and finding the maximum value in a matrix.

Two-dimensional arrays are powerful tools for representing structured data. They are widely used in mathematics (matrices), computer graphics (images), and engineering applications (grids and tables). By mastering 2D arrays, students gain the ability to handle more complex problems involving multidimensional data. This lab provided practical experience that strengthens programming skills and prepares students for advanced topics like dynamic memory allocation and algorithm design.